



Biotechnology

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Using living systems to develop products and technologies

Biotechnology uses living organisms, cells, or biological systems to develop products and technologies for medicine, agriculture, and industry.

● DEFINITION

The use of biological systems, living organisms, or their derivatives to develop or create products and processes for specific uses.

● WHY IT MATTERS

Biotechnology underpins modern medicine, agriculture, and industrial production — from life-saving drugs to disease-resistant crops.

● WORKING PRINCIPLE



Biotechnologists manipulate genetic material or cellular processes in organisms — bacteria, yeast, plants, or animal cells — to produce useful compounds like insulin, enzymes, or improved crop traits.

● QUICK FACTS

- ▶ Insulin was the first genetically engineered protein approved for medical use, in 1982
- ▶ CRISPR gene editing technology won the 2020 Nobel Prize in Chemistry

● KEY TERMS

Recombinant DNA Gene editing
Fermentation GMO

● CORE CONCEPTS

- ▶ Genetic engineering
- ▶ Fermentation technology
- ▶ Recombinant DNA technology
- ▶ Cell culture techniques

● APPLICATIONS

- ▶ Insulin and vaccine production
- ▶ Genetically modified crops
- ▶ Industrial enzyme production
- ▶ Gene therapy development

● ADVANTAGES

- ▶ Enables production of complex biological drugs
- ▶ Improves agricultural yield and resilience
- ▶ Drives major medical breakthroughs

CAREER OPPORTUNITIES

Biotechnologist, Genetic engineer, Bioprocess engineer, Research scientist

RESEARCH AREAS

Gene editing (CRISPR), Synthetic biology, Cell and gene therapy

● REAL-LIFE EXAMPLES

- ▶ Insulin produced by genetically engineered bacteria
- ▶ Disease-resistant GMO crops
- ▶ Enzyme-based laundry detergents

● LIMITATIONS

- ▶ Public perception concerns around genetic modification
- ▶ Complex and costly regulatory approval
- ▶ Ethical considerations in genetic engineering

INDUSTRY APPLICATIONS

Pharmaceuticals, Agriculture, Industrial manufacturing, Environmental remediation

FUTURE SCOPE

Rapid advances in gene editing tools like CRISPR are expanding biotechnology's reach into precision medicine and agriculture.

● CURRENT TRENDS

Explosive growth in CRISPR-based gene editing applications across medicine and agriculture.

● SUMMARY

Biotechnology harnesses living systems to create medicines, improved crops, and industrial products, with gene editing driving its fastest growth.

